FINITENESS OF COMPACTLY SUPPORTED COHOMOLOGY OVER AN ALGEBRAICALLY CLOSED FIELD

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Theorem: for a separated morphism of finite presentation of quasi-compact quasi-separated schemes and a Noetherian ring Lambda, Rf_! sends D^+_{tors, c} (or D_c if Lambda is torsion) into D^+_{tors, c}(Y) (or D_c(Y)), i.e. Rf_!K has constructible cohomology sheaves (more-etale, Theorem theorem-constructible-shriek, canonical tag 0GL0). cited at proof line 4997. The named theorem giving constructibility of Rf_!K, of which the finiteness of H^i_c is stated to be an immediate consequence.
- Section defining compactly supported cohomology (more-etale, Section section-compactly-supported-cohomology, canonical tag 0GJY). cited at proof lines 4998-4999. The definition of the compactly supported cohomology groups H^i_c(X, K) whose finiteness is being proven.

Source reference. Original theorem 0GL0: sources/stacks-chow/tags.txt, source lines 20705–20705. Cited at proof line 4997: the statement is an immediate consequence of this theorem.

Source reference. Original section 0GJY: sources/stacks-chow/tags.txt, source lines 20668–20668. Cited at proof lines 4998-4999: the definition of H^i_c used to read off finiteness.

Lemma 0.2. *Let k be an algebraically closed field. Let X be a finite type separated scheme over k. Let Λ be a Noetherian ring. Let K be an object of $D^+_{tors,c}(X_{étale}, \Lambda)$ or of $D_c(X_{étale}, \Lambda)$ in case Λ is torsion. Then $H^i_c(X, K)$ is a finite Λ-module for all $i \in \mathbf{Z}$.*

Proof. Immediate consequence of Theorem 0GL0 and the definition of compactly supported cohomology in Section 0GJY. □

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